

Exercício 6
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No.	Time	Source	Destination	Protocol	Length	Info
37	131.457665603	10.0.2.15	185.125.190.122	NTP	230	NTP Version 4, client, NTS
38	131.652367297	185.125.190.122	10.0.2.15	NTP	230	NTP Version 4, server, NTS
39	132.635184832	10.0.2.15	91.189.91.113	NTP	230	NTP Version 4, client, NTS
40	132.765792632	91.189.91.113	10.0.2.15	NTP	230	NTP Version 4, server, NTS
41	134.423272702	10.0.2.15	185.125.190.121	NTP	230	NTP Version 4, client, NTS
42	134.618014153	185.125.190.121	10.0.2.15	NTP	230	NTP Version 4, server, NTS
43	136.717679521	fd17:625c:f037:2:a0::	ff02::fb	MDNS	180	Standard query 0x8000 PTR _ftp._tcp.local, "QM" question PTR _nfs._tcp.local, "QM" question PTR _afpovertcp
44	136.717941272	10.0.2.15	224.0.0.251	MDNS	160	Standard query 0x8000 PTR _ftp._tcp.local, "QM" question PTR _nfs._tcp.local, "QM" question PTR _afpovertcp
45	158.885956241	10.0.2.15	8.8.4.4	DNS	100	Standard query 0x8235 A connectivity-check.ubuntu.com OPT
46	158.905022011	8.8.4.4	10.0.2.15	DNS	190	Standard query response 0x0235 A connectivity-check.ubuntu.com A 185.125.190.100 A 91.189.91.97 A 91.189.91
47	158.905985150	10.0.2.15	185.125.190.100	TCP	74	39210 -> 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=363265966 TSecr=0 WS=512
48	159.100465844	185.125.190.100	10.0.2.15	TCP	60	80 -> 39210 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
49	159.100484375	10.0.2.15	185.125.190.100	TCP	54	39210 -> 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
50	159.100965177	10.0.2.15	185.125.190.100	HTTP	141	GET / HTTP/1.1
51	159.101275089	185.125.190.100	10.0.2.15	TCP	60	80 -> 39210 [ACK] Seq=1 Ack=88 Win=65535 Len=0
52	159.296320793	185.125.190.100	10.0.2.15	HTTP	239	HTTP/1.1 204 No Content
53	159.296321292	185.125.190.100	10.0.2.15	TCP	60	80 -> 39210 [FIN, ACK] Seq=180 Ack=88 Win=65535 Len=0
54	159.296348393	10.0.2.15	185.125.190.100	TCP	54	39210 -> 80 [ACK] Seq=88 Ack=180 Win=64055 Len=0
55	159.296695925	10.0.2.15	185.125.190.100	TCP	54	39210 -> 80 [FIN, ACK] Seq=88 Ack=187 Win=64055 Len=0
56	159.296895438	185.125.190.100	10.0.2.15	TCP	60	80 -> 39210 [ACK] Seq=187 Ack=89 Win=65535 Len=0
57	164.339959653	PCSSystemtec_of:5a::	52:54:00:12:35:00	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15

No.	Time	Source	Destination	Protocol	Length	Info
18	67.063764961	185.125.190.122	10.0.2.15	NTP	230	NTP Version 4, server, NTS
19	68.535180225	10.0.2.15	91.189.91.113	NTP	230	NTP Version 4, client, NTS
20	68.605182176	91.189.91.113	10.0.2.15	NTP	230	NTP Version 4, server, NTS
21	69.61885872	PCSSystemtec_of:5a::	52:54:00:12:35:00	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
22	69.628328304	52:54:00:12:35:00	PCSSystemtec_of:5a::	ARP	64	10.0.2.2 is at 52:54:00:12:35:00
23	69.84386286	10.0.2.15	185.125.190.121	NTP	230	NTP Version 4, client, NTS
24	70.638345752	185.125.190.121	10.0.2.15	NTP	230	NTP Version 4, server, NTS
25	103.892406535	10.0.2.15	8.8.4.4	DNS	83	Standard query 0x9df2 A neverssl.com OPT
26	103.892736708	10.0.2.15	8.8.4.4	DNS	83	Standard query 0x5ab4 AAAA neverssl.com OPT
27	103.911402428	8.8.4.4	10.0.2.15	DNS	99	Standard query response 0x9df2 A neverssl.com A 34.223.124.45 OPT
28	103.911402818	8.8.4.4	10.0.2.15	DNS	111	Standard query response 0x5ab4 AAAA neverssl.com AAAA 2600:1f13:37c:1400:ba21:7165:5fc7:736e OPT
29	103.913047582	10.0.2.15	34.223.124.45	TCP	74	43466 -> 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=116911412 TSecr=0 WS=512
30	103.913047582	10.0.2.15	34.223.124.45	TCP	74	43466 -> 80 [SYN, ACK] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=116911412 TSecr=0 WS=512
31	105.119571911	34.223.124.45	10.0.2.15	TCP	60	80 -> 43466 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
32	105.119643608	10.0.2.15	34.223.124.45	TCP	54	43466 -> 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
33	129.712330499	10.0.2.15	91.189.91.112	NTP	230	NTP Version 4, client, NTS
34	129.842788484	91.189.91.112	10.0.2.15	NTP	230	NTP Version 4, server, NTS
35	130.397058477	10.0.2.15	185.125.190.123	NTP	230	NTP Version 4, client, NTS
36	130.591997719	185.125.190.123	10.0.2.15	NTP	230	NTP Version 4, server, NTS
37	131.457665603	10.0.2.15	185.125.190.122	NTP	230	NTP Version 4, client, NTS
38	131.652367297	185.125.190.122	10.0.2.15	NTP	230	NTP Version 4, server, NTS

Frame 35: Packet, 230 bytes on wire (1840 bits), 230 bytes captured (1840 bits) on interface enp0s3, Ethernet II, Src: PCSSystemtec_of:5a:5f (08:00:27:0f:5a:5f), Dst: 52:54:00:12:35:00 (52:54:00:12:35:00)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 185.125.190.123

User Datagram Protocol, Src Port: 56463, Dst Port: 123

Network Time Protocol (NTP Version 4, client)

0000 52 54 00 12 35 00 00 00 27 0f 5a 5f 08 00 45 00 RT=5...Z...E
0010 00 d8 69 97 40 00 40 11 4c 76 0a 00 02 0f b9 7d ...1.0.0.LV...
0020 be 7b dc 8f 00 7b 00 c4 84 dd 23 00 00 20 00 00 {...}...#...
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...
0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...
0050 00 00 de b1 13 e9 71 d6 7a e7 01 04 00 24 b7 f5q.Z...S...
0060 2b db 08 2e e6 38 75 44 b7 30 68 ba af 92 ac 92 +...8y0.0h...
0070 31 25 47 2f 59 93 b9 55 27 0b e9 aa 0b 8f 02 04 1%G/Y..U.....
0080 00 44 5d 12 60 90 86 f1 c8 9d 01 dc 25 d5 5b f5 [...].....%[...
0090 42 5a 52 00 29 24 a9 73 e3 e2 ba ee a9 6d f8 66 B.R)S.S.....f...
00a0 5a cb 87 58 d6 07 35 f7 ed ce 4f a9 4a 6d 56 12 Z..X..5...0.JmV...
00b0 29 92 2a 2b b3 0f 91 c1 cd 64 73 db 0d 0c 69 ae).+*....ds...1...
00c0 50 35 04 04 00 24 00 0c 00 10 c7 0e e0 1e 2e 1a P5..\$......
00d0 21 06 d5 51 41 24 b7 ff 5a 63 5a bc 76 bf 60 f9 .f.QAS..Zc2.v...
00e0 9f 09 b8 e9 e9 80 13

A captura realizada no Wireshark permitiu observar todo o ciclo de uma conexão TCP. Inicialmente ocorreu o estabelecimento da conexão por meio do Three-Way Handshake, com a troca dos segmentos SYN, SYN/ACK e ACK (pacotes 47, 48 e 49). Após a conexão ser estabelecida, foi realizada uma requisição HTTP GET ao servidor (pacote 50), que respondeu com uma mensagem HTTP 204 No Content (pacote 52). Em seguida, observou-se o encerramento da conexão TCP através da troca dos segmentos FIN/ACK e ACK (pacotes 53 a 56), finalizando a comunicação entre cliente e servidor de forma ordenada.